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ORIGINAL STUDY

Anomaly Detection in Electrical Systems Using Machine Learning and Statistical Analysis

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ABSTRACT

The reliability of electrical systems is essential for uninterrupted power supply across residential, industrial, and commercial sectors. This research utilizes the various techniques of ML in identifying all sorts of abnormal electrical power variables, including voltage fluctuations, line faults, and frequency variations that emanate from several causes like equipment failures and environmental ones. Voltage and current measurements were pre-processed along with other additional parameters such as frequency and THD, then visualized to assure its quality and relevance for training the ML models. The work implemented the algorithms of Random Forest, Decision Tree, XGBoost, SVM, and Logistic Regression-the Random Forest Classifier performed best for improving results with the best accuracy 99%, precision of 99%, and recall 99%. Random Forest yielded much higher anomaly detection with better generalization after an appropriate avoidance of the overfit to data via effective ensemble. Performance measures obtained against the confusion matrix will really underscore that it makes pretty good generalizations to an independent/unseen data distribution also. This research underlines the potential of data-driven techniques to enhance fault detection, improve predictive maintenance, and ensure grid reliability. Future work may focus on real-time deployment and scalability to integrate such systems into operational grid environments for proactive anomaly management.

Keywords: Electrical fault detection, Machine learning, Random forest, Predictive maintenance

1. Introduction and literature review

Reliability in electrical systems is one of the key foundations of modern civilization, being fundamental to all sectors of activity, including residential, commercial, and industrial.^{1–3} Electricity provides a continuous source from residences and hospitals to manufacturing plants and data centers that keep the pulse of human life going without any operational downtimes, thus ensuring economic stability.^{4,5} Disruption in the grid results in a snowballing effect: disrupted operations, financial loss, and even safety hazards. While the demand for electrical systems is increasing, making them more complex to meet that demand, ensuring stability and efficiency becomes a

big challenge. Intrinsically, these systems are prone to a variety of disturbances and anomalies.^{6–9} The general problems of supply line distortions, mechanical failures, voltage fluctuations, and frequency variations may arise due to a variety of causes such as aging infrastructure, malfunctioning equipment, environmental factors like storms or lightning strikes, and even malicious activities like cyber-attacks.

These faults, if left unattended without immediate remediation, have the potential to affect critical system integrity and lead to systemwide blackouts.^{10–13} Thus, detection and resolution of faults within electrical systems has emerged as one of the high-priority areas for research and development.^{14,15} Many traditional detection methods based

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on manual observation and rule-based algorithms are starting to fail due to the enormous scale and complex nature of large-scale modernized grids.^{16,17} These methods may be insufficient to recognize subtle patterns and predict possible failures, thus systems are bound to become merely reactive.^{18,19} The field, in turn, has increasingly moved toward advanced technological solutions, especially the power of data-driven techniques and machine learning.^{20–22} Such large volumes of operational data analyses can enable machine learning models to identify anomalies with greater precision, adapt to evolving patterns, and even predict faults before they happen. This proactive approach contributes not only to enhanced resilience for electrical systems but also to reduced downtime and maintenance costs. While the call is increasingly raised for smarter and dependable grids, the development of sophisticated fault detection systems with intelligent analytics and automation is still paramount in research and practice.

Electrical systems encompass a wide range of power quality anomalies, from voltage fluctuations and line faults to frequency variations, due to equipment failures or environmental factors or even malicious activities. Traditional detection methods for such distortions rely on manual inspection and rule-based algorithms, which may fall short in handling the modern grid system in terms of its scale and complexity. It follows that there is an increasing interest in advanced data-driven techniques, especially machine learning models, for fault detection in an Electrical system in a more accurate and efficient way.

There are a number of studies that have assessed the suitability of ML models in detecting supply line distortions in the electric grid. The range of techniques employed in these studies ranges widely—from classical supervised learning algorithms such as Decision Trees, Random Forest, and Support Vector Machines (SVM),^{23,24} to Neural Networks—on one hand,^{25,26} while unsupervised learning techniques include clustering and anomaly detection. Each of these methods has its strengths and weaknesses regarding accuracy, scalability, and interpretability and hence useful for different facets of the system. Decision Trees are among the most common approaches to supervised learning because of their simplicity and interpretability.²⁷ A decision tree works by recursively partitioning the input space depending on the values of features. Therefore, they are suited for picking out pattern's indicative of anomalies. The ensemble learning technique comprising of many decisions trees—Random Forest has also yielded very promising results in view of improving the accuracy and robustness of detection.

Vu Le, 2020²⁸ described an arc faults study in different power electronics loads, with the extracted features tested for computational efficiency. After they were extracted, the ensemble machine learning algorithm effectively detected arc faults outperforming state-of-the-art methods and other machine learning methods. In this work, ensemble techniques, namely, bagging, boosting, and stacking were considered in order to evaluate EML algorithms for accuracy, precision, and recall. Therefore, the stacking ensemble outperformed all other techniques and yielded a low-cost hardware arc fault detector with minimum latency. It showed versatility and reliability over a wide range of load types further tested with transient start-ups and parallel connections.

Overall, the literature underscores the potential of ML models in enhancing the detection capabilities of faults in Electrical systems. By leveraging advanced data analytics techniques, utilities can gain deeper insights into grid operation and proactively address potential issues, thereby improving grid reliability and resilience. However, challenges remain in terms of data quality, feature selection, and model interpretability, which necessitate further research and development efforts in this domain.

In this paper, some efforts have been put together to have good results. Voltage and current patterns of individual lines are monitored and Multiple other parameters were collected which are mentioned in the later section of Data Processing, and decisions have been made on certain anomalies. The results obtained are quite satisfactory.

2. Data preprocessing and visualization

2.1. Data collection and preprocessing

The study commenced by collecting data comprising current and voltage measurements for all the phases within the electrical systems and multiple other parameters such as frequency, THD and etc.²⁹ The heat map of the parameter's correlation is given below in Fig. 1.

Correlation heat map highlights how different input are changing with the change in others input and how they proportional to each other.

Before feeding into the model the data was preprocessed visualized thoroughly, which is the first and foremost step to develop any ML model. Doing this also helps us with the model selection. It helps us which model to use and which model give us the best results given the parameters of the model. In this Project The data was collected from Kaggle website.²⁹

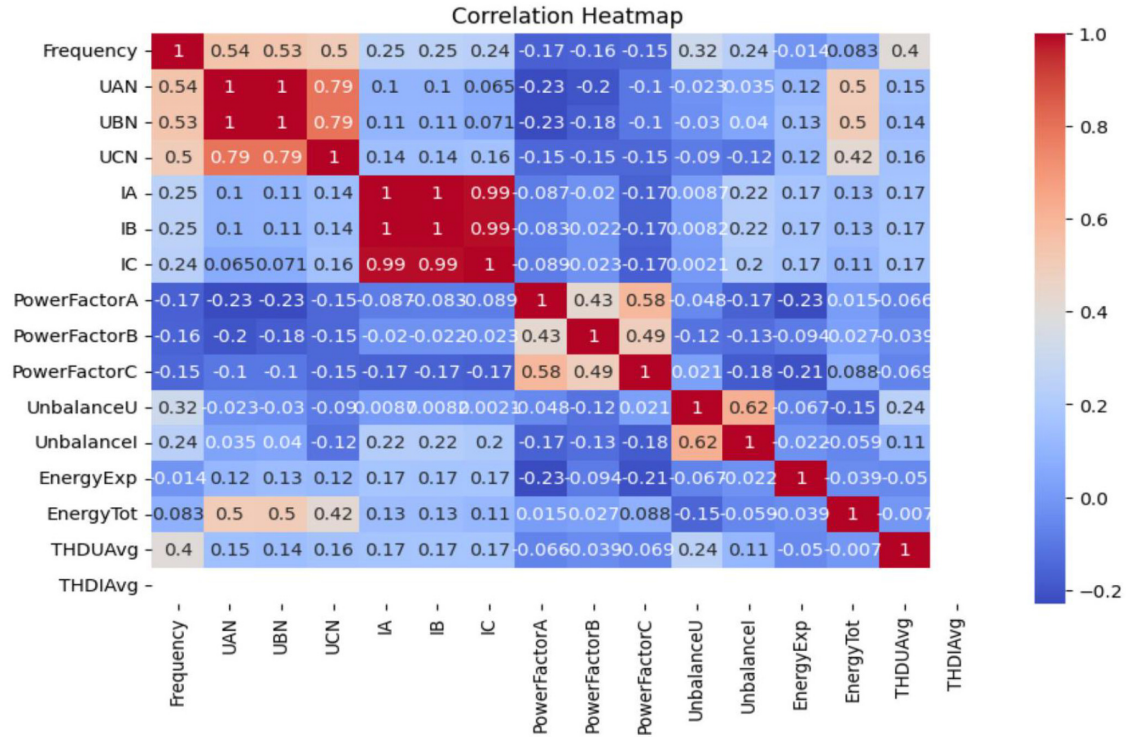


Fig. 1. Features correlation heat map.

Processed and visualized using different techniques such as removing outliers, scaling and standardization and feature engineering.

2.2. Train and test split

In the machine learning pipeline, selecting appropriate input features, such as voltage and current measurements, and output features, like the status of the anomalies detected, is fundamental for model training. Once these features are identified, the dataset is divided into two subsets: a training set and a testing set. The training set, comprising the majority of the data (typically 70–80%), is used to train the model on patterns and relationships within the data. Meanwhile, the testing set, constituting the remaining portion (20–30%), remains untouched during the training process and serves as a benchmark to evaluate the model's performance on unseen data. This partitioning ensures that the model's effectiveness and generalization ability are properly assessed, guarding against overfitting or underfitting.

So, input and output features were selected, and the data was partitioned into training and testing sets to further implement the Machine Learning Models.

2.3. Model implementation

The subsequent phase involved the implementation of multiple Machine Learning models, including Random Forest, Decision Tree, XGBoost, Support Vector Machine (SVM), and Logistic Regression.

Implementation of the machine learning model involves selection, training, and evaluation of the predictive algorithm for a given dataset. Given that the dataset is cleaned and prepared into training and test sets, a model now needs to be trained based on the data of training and selection of input features previously chosen. In training the model, its parameters are modified in a way aimed at reducing or bringing closer the value of output forecasted to that actually measured. It can be implemented based on the nature of the problem and the characteristics of the data by various machine learning algorithms, such as Decision Trees, Random Forest, XGBoost, SVM, and Logistic Regression. Here we have applied almost all the models to see which one works best.

After training, the model's performance is evaluated using the testing data, measuring metrics such as accuracy, precision, recall, and F1-score. This evaluation helps assess how well the model generalizes to unseen data and provides insights into its effectiveness in making predictions. Additionally, techniques

like hyperparameter tuning and cross-validation may be employed to optimize the model’s performance further and ensure its robustness across different datasets.

• Random Forest Classifier

Random Forest model was trained on the training data. There were observed compatible results exhibiting a training accuracy of an impressive 99%. The results are shown below in Table 1.

The reason why this model was used and it gave the highest results is that random forest can reduce overfitting largely by getting results from its multiple decision trees. Random Forest Algorithm is made up of a collection of decision trees. And it takes results from every tree, which makes it having less over fitting and under fitting chances. The chart is given below in Fig. 2.

Now the results of the system has been derived and the confusion matrix of the results is given below in Fig. 3.

It can be seen from the confusion matrix that a few of the predictions are going wrong direction otherwise the model is performing quite well.

Table 1. Assessment metrics results.

Assessment Metric	Result (%)
Accuracy	99
Precision	98
Recall	99

Once the model is trained and evaluated satisfactorily, it can be deployed to make predictions on new, unseen data, thereby providing valuable insights and aiding decision-making processes in various domains, including healthcare, finance, and engineering. Model implementation thus forms a critical step in the machine learning workflow, bridging the gap between data analysis and real-world applications.

3. Results and discussion

The results in Table 2 indicate that the Random Forest Classifier outperformed all other models, achieving the highest accuracy (99%) along with strong precision (98%) and recall (99%). This highlights its robustness in detecting electrical anomalies while avoiding overfitting due to ensemble learning. XGBoost followed closely with 97% accuracy, demonstrating its efficiency in handling structured data. Decision Tree performed well but showed slightly lower generalization, while SVM and Logistic Regression yielded lower accuracy, indicating their limitations in complex anomaly detection tasks. Overall, Random Forest emerged as the most reliable model for electrical fault classification.

Among all the ML models that were tested within this research, the Random Forest Classifier performed the best: 99% for accuracy, precision, and recall. This clearly shows the strength and adaptiveness of

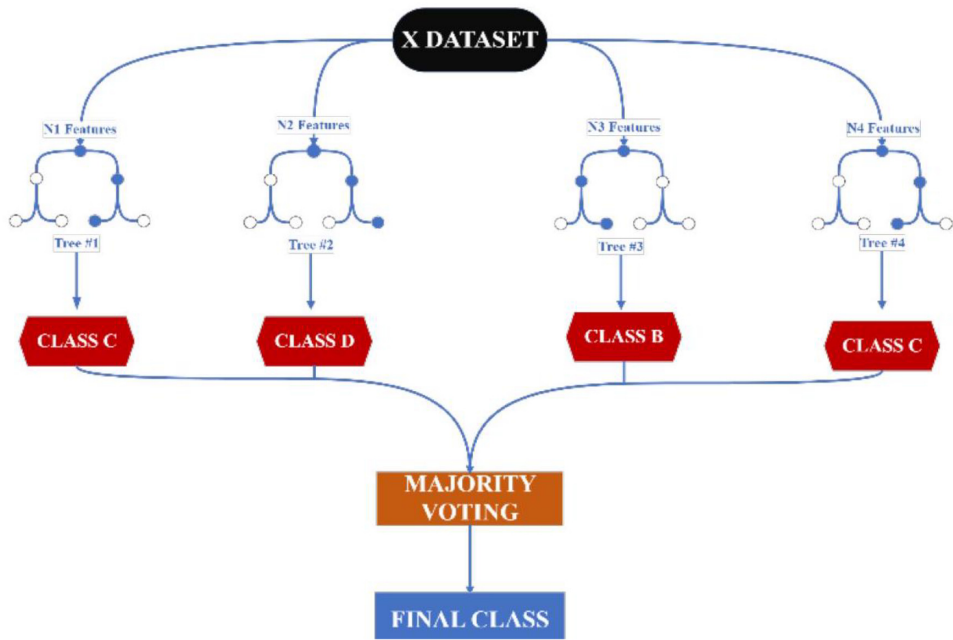


Fig. 2. Workflow chart.

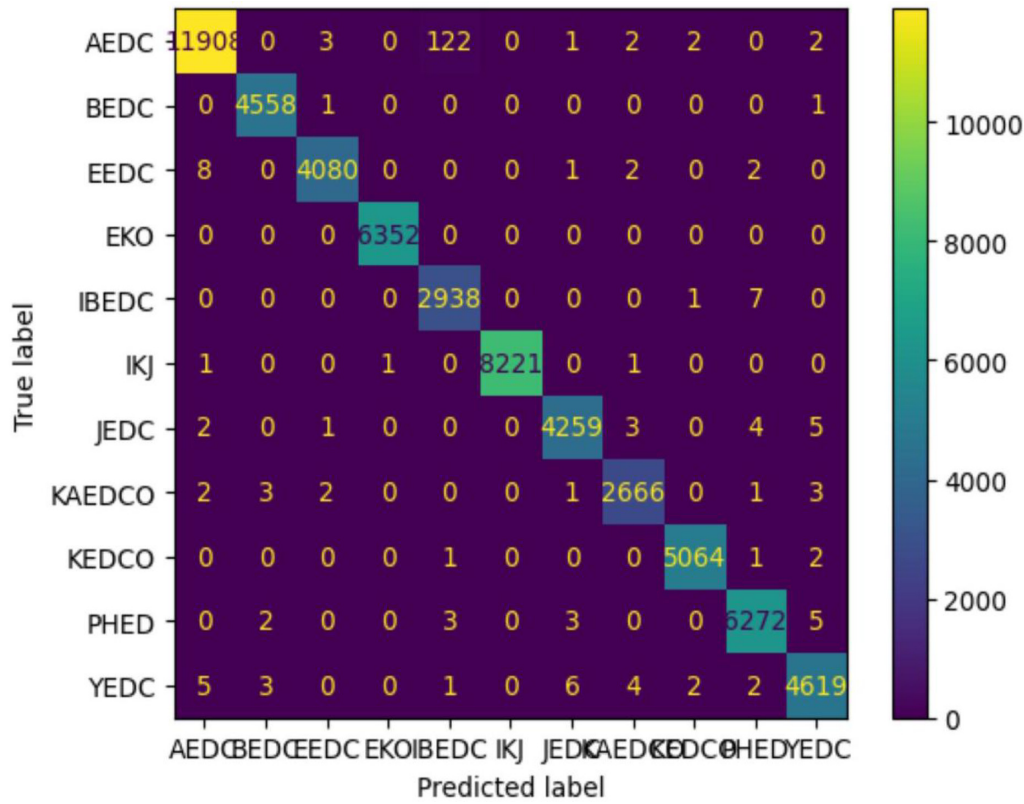


Fig. 3. Random forest confusion matrix.

Table 2. Classification model results.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	99	98	99	98.5
Decision Tree	94	92	94	93
XGBoost	97	95	96	95.5
SVM	89	87	88	87.5
Logistic Regression	85	83	84	83.5

Random Forest as a predictive tool for detecting and classifying anomalies in electrical systems. It is the ensemble nature of the Random Forest algorithm, which pools decisions from multiple decision trees, that makes it really strong in avoiding overfitting without sacrificing high generalization capability. This makes it especially suitable for dealing with complex datasets, such as those involving electrical faults, where there is a variety of patterns and even minute anomalies that must be correctly identified. Analysis of the confusion matrix also proved that the classifier is reliable with few wrong predictions, thus proving to be a potent method for anomaly detection.

While these results are promising, several challenges must be overcome to translate such findings into a real-time operational grid environment. This means integrating the Random Forest model with monitoring systems that can handle huge volumes of inbound live data from various sources and pro-

cess them in near real time. The implementation requires low-latency data transfer, efficient preprocessing pipelines, and model updates in real time, adapting to the emergence of new patterns. Besides, scalability is considered one of the most important challenges in electrical grids, as these usually serve broad areas with complicated network topological structure. Continuously performing model validation and updating are necessary, too, because, under various situations, such as variations in the load demand, aging of equipment, or environmental conditions, the performance maintained by the model is expected to be of high standard.

Another important aspect is the economic feasibility of deploying such machine learning-based systems in practical scenarios. While the Random Forest Classifier has excellent detection capabilities, the computational resources for training and real-time decision-making have to be weighed against

the cost-benefit ratio. The implementation of scalable solutions, such as edge computing or cloud-based architectures, could offer a pathway to overcoming these challenges. Besides, there are regulatory and operational considerations that also determine the success of such systems, such as adherence to grid standards and integration with existing fault management protocols.

4. Conclusion

In a nutshell, this paper presented the development concerning an advanced model using machine learning in order to detect and classify anomalies in electrical systems. It considered for careful analysis crucial parameters involving voltage as well as current measurements of each line and between them, frequency, THD, power factor, energy input, and energy output to detect an anomaly with high precision, indicating an impending electrical fault. With extensive data preprocessing and tight model implementation, the Random Forest algorithm turned out to be the most powerful, giving an exceptional performance of 99% in terms of accuracy, precision, and recall. Clearly, this underlines the strength and flexibility of machine learning techniques in the improvement of fault detection to enable predictive maintenance. This approach is opening the path to more effective and proactive maintenance activities, providing utilities with actionable information on system behavior and with early warning against impending problems. Additionally, the potential of data-driven methods was demonstrated to support the advancement in reliability, resilience, and operational efficiency for modern electrical systems, marking another important step towards the integration of artificial intelligence within the domain of grid management and fault diagnostics.

References

1. Zhao C, Zhang B, Zheng Y, Huang S, Yan T, Liu X. Hybrid battery thermal management system in electrical vehicles: A review. *Energies (Basel)*. 2020;13(23):6257.
2. Bafrani AA, Rezazade A, Sedighzadeh M. Robust electrical reserve and energy scheduling of power system considering hydro pumped storage units and renewable energy resources. *J Energy Storage*. 2022;54:105310. doi: <https://doi.org/10.1016/j.est.2022.105310>.
3. Bhat D, Muench S, Roellig M. Application of machine learning algorithms in prognostics and health monitoring of electronic systems: A review. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*. 2023;4:100166. doi: <https://doi.org/10.1016/j.eprime.2023.100166>.
4. Kowsar A, Naima F, Rana MT, Haque N, Alam F. Techno-economic study of a photovoltaic power plant besides the railway track for rural uses in Bangladesh. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*. 2023;5:100229. doi: <https://doi.org/10.1016/j.eprime.2023.100229>.
5. Shakouri H, Pandey S, Rahmatian F, Paaso EA. Does the increased electricity consumption (provided by capacity expansion and/or reliability improvement) cause economic growth?. *Energy Policy*. 2023;182:113704. doi: <https://doi.org/10.1016/j.enpol.2023.113704>.
6. Bonci A, Longhi S, Nabissi G. Fault diagnosis in a belt-drive system under non-stationary conditions. An industrial case study. In *2021 IEEE Workshop on Electrical Machines Design, Control and Diagnosis (WEMDCD)*. 2021;260–265. doi: [10.1109/WEMDCD51469.2021.9425680](https://doi.org/10.1109/WEMDCD51469.2021.9425680).
7. Patra AK, Rath D, Kar SK. Performance evaluation of grid connected photovoltaic system using novel HTLB-WOA regulated VPOTADF controller. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*. 2023;3:100098. doi: <https://doi.org/10.1016/j.eprime.2022.100098>.
8. Alawee WH, Basem A, and Al-Haddad LA. Advancing biomedical engineering: Leveraging Hjorth features for electroencephalography signal analysis. *J Electr Bioimpedance*. 2023;14(1):66–72. doi: [10.2478/joeb-2023-0009](https://doi.org/10.2478/joeb-2023-0009).
9. Shijer SS, Jassim AH, Al-Haddad LA, Abbas TT. Evaluating electrical power yield of photovoltaic solar cells with k-Nearest neighbors: A machine learning statistical analysis approach. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*. 2024;9:100674. doi: <https://doi.org/10.1016/j.eprime.2024.100674>.
10. Ewert P, Kowalski CT, Orlowska-Kowalska T. Low-cost monitoring and diagnosis system for rolling bearing faults of the induction motor based on neural network approach. *Electronics (Basel)*. 2020;9(9):1334.
11. Al-Haddad LA, Shijer SS, Jaber AA, Al-Ani ST, Al-Zubaidi AA, Abd ET. Application of AdaBoost for stator fault diagnosis in three-phase permanent magnet synchronous motors based on vibration–current data fusion analysis. *Electrical Engineering*. 2024. doi: [10.1007/s00202-023-02195-6](https://doi.org/10.1007/s00202-023-02195-6).
12. Al-Haddad LA, Jaber AA, Hamzah MN, Fayad MA. Vibration-current data fusion and gradient boosting classifier for enhanced stator fault diagnosis in three-phase permanent magnet synchronous motors. *Electrical Engineering*. 2023. doi: [10.1007/s00202-023-02148-z](https://doi.org/10.1007/s00202-023-02148-z).
13. Al-Haddad LA, Ibraheem L, EL-Seesy AI, Jaber AA, Al-Haddad SA, Khosrozadeh R. Thermal heat flux distribution prediction in an electrical vehicle battery cell using finite element analysis and neural network. *Green Energy and Intelligent Transportation*, p. 100155, 2024. doi: <https://doi.org/10.1016/j.geits.2024.100155>.
14. Glowacz A. Thermographic fault diagnosis of electrical faults of commutator and induction motors. *Eng Appl Artif Intell*. 2023;121:105962. doi: <https://doi.org/10.1016/j.engappai.2023.105962>.
15. Bai K, Sindhu V, Haque A. Fault ride through approach for grid-connected photovoltaic system. *e-Prime - Advances in Electrical Engineering, Electronics and Energy*. 2023;5:100232. doi: <https://doi.org/10.1016/j.eprime.2023.100232>.
16. Wadi M. Fault detection in power grids based on improved supervised machine learning binary classification. *Journal of Electrical Engineering*. 2021;72(5):315–322. doi: [10.2478/jee-2021-0044](https://doi.org/10.2478/jee-2021-0044).
17. Hyun D, Kang T-J, Kim J. Fault diagnosis of belt driving power transfer facility using vibration analysis. *Journal of Electrical Engineering & Technology*. 2021;16(4):2257–2264. doi: [10.1007/s42835-021-00711-x](https://doi.org/10.1007/s42835-021-00711-x).
18. Tang J, Liang C, Wang Y, Lu S, Zhou J. A stator fault diagnosis method based on the offline motor parameter measurement

- for PMSM. *World Electric Vehicle Journal*. 2021;12(4). doi: [10.3390/wevj12040248](https://doi.org/10.3390/wevj12040248).
19. Moosavi SS, Djerdir A, Ait-Amirat Y, Khaburi DA. ANN based fault diagnosis of permanent magnet synchronous motor under stator winding shorted turn. *Electric Power Systems Research*. 2015;125:67–82. doi: <https://doi.org/10.1016/j.epsr.2015.03.024>.
 20. Pietrzak P, Wolkiewicz M. Machine learning-based stator current data-driven PMSM stator winding fault diagnosis. *Sensors*. 2022;22(24). doi: [10.3390/s22249668](https://doi.org/10.3390/s22249668).
 21. Long J, Chen Y, Yang Z, Huang Y, Li C. A novel self-training semi-supervised deep learning approach for machinery fault diagnosis. *Int J Prod Res*. Dec. 2023;61(23):8238–8251. doi: [10.1080/00207543.2022.2032860](https://doi.org/10.1080/00207543.2022.2032860).
 22. Mejbel BG, Sarow SA, Al-Sharify MT, Al-Haddad LA, Ogaili AAF, Al-Sharify ZT. A data fusion analysis and random forest learning for enhanced control and failure diagnosis in rotating machinery. *Journal of Failure Analysis and Prevention*. 2024. doi: [10.1007/s11668-024-02075-6](https://doi.org/10.1007/s11668-024-02075-6).
 23. Caballero EF, Lobo DM, Di Vaio MV, Silva ECCM, Ritto TG. Support vector machines applied to torsional vibration severity in drill strings. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*. 2021;43(8):386. doi: [10.1007/s40430-021-03089-z](https://doi.org/10.1007/s40430-021-03089-z).
 24. Yaman O, Yol F, Altinors A. A sound based method for fault classification with support vector machines in UAV motors. *International Journal of Data Science and Applications*. 2021;4(1):5–10, [Online]. Available: <https://dergipark.org.tr/en/pub/joindata/issue/83159/1437091>
 25. Al-Haddad LA *et al*. Enhancing building sustainability through aerodynamic shading devices: An integrated design methodology using finite element analysis and optimized neural networks. *Asian Journal of Civil Engineering*, 2024. doi: [10.1007/s42107-024-01047-3](https://doi.org/10.1007/s42107-024-01047-3).
 26. Al-Haddad LA, Alawee WH, Basem A. Advancing task recognition towards artificial limbs control with ReliefF-based deep neural network extreme learning. *Comput Biol Med*. 2023, 107894. doi: <https://doi.org/10.1016/j.compbiomed.2023.107894>.
 27. Yao S, Kronenburg A, Shamooni A, Stein OT, Zhang W. Gradient boosted decision trees for combustion chemistry integration. *Applications in Energy and Combustion Science*. 2022;11:100077. doi: <https://doi.org/10.1016/j.jaecs.2022.100077>.
 28. Le V, Yao X, Miller C, Tsao B-H. Series DC arc fault detection based on ensemble machine learning. *IEEE Trans Power Electron*. 2020;35(8):7826–7839. doi: [10.1109/TPEL.2020.2969561](https://doi.org/10.1109/TPEL.2020.2969561).
 29. Jose J. Electrical fault detection and classification. Kaggle Dataset. Accessed: Nov. 18, 2024. [Online]. Available: <https://www.kaggle.com/code/jaison14/electrical-fault-detection-and-classification>.